QP CODE: 25900141

MG U-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

Second Semester

Discipline Specific Core Course (DSC) - MG2DSCMBG101 - MICROBIAL PHYSIOLOGY AND FOOD MICROBIOLOGY

(2024 Admission onwards)

PART A

1. Lag phase (1mark)
2. General purpose medium (1mark)
3. Ammonia (1mark)
4. Differential medium (1mark)
5. Aerobes (1mark)
6. Fermentation (1mark)
7. Nutrient content (1mark)
8. All the above (1mark)
9. Bacterial endospores (1mark)
10. Formaldehyde (1mark)

PART B

11. Binary fission (1mark)
12. EMB agar/ Mac Conkey agar/ Mannitol salt agar/Salmonella Shigella agar/DCA/TCBS etc (any one) (1mark)
13. Synthetic or defined media (1mark)
14. Hydrogen ion concentration /water activity / nutrient content / oxidation reduction potential (any one) (1mark)
15. Carbon dioxide, ethanol, lactic acid, lactoperoxidase, low P^H, low O-R potential, low water activity, maillard reaction, organic acids, oxygen, ozone,phenols, salt, smoking etc (any one) (1mark)

PART C

16. False (1mark)
17. False (1mark)
18. False (1mark)
19. True (1mark)
20. False (1mark)

PART D

21. Microbes that use light as a source of energy and organic compounds as the main source of carbon. (1mark)

22. MacConkey agar is a selective and differential culture medium for bacteria, selectively isolate gram-negative bacteria and differentiate them based on lactose fermentation. (1mark)

23. Mac Conkey agar/ EMB agar/ Blood agar etc (any one) (1mark)

24. Nutrient agar is a solid medium supporting growth of a wide range of non-fastidious organisms. (1mark)

25. Microbes that grow in the absence of oxygen. (1mark)

26. Deterioration of food due to microbial activity, rendering it undesirable or unsafe for consumption. (1mark)

27. *Saccharomyces cerevisiae* (1mark)

28. Bacteria, virus, fungus, parasites etc (1mark)

29. Generally Recognized as Safe. (1mark)

30. Shell of eggs, skin of vegetables, skin of fruits , shell of nuts, shell of nutmeats, husk of rice etc (any two) (1mark)

31. Used to sterilize some food products such as spices, dried vegetables, used as a preservative and fumigant in the food industry, especially in the storage room. (1mark)

32. Solar drying is the cheapest method of drying that evaporates moisture from foods. (1mark)

PART E

33. Bacteria are classified based on their pH requiremnts into three main groups: acidophiles, neutrophiles, and alkaliphiles.
(1.5mark). Acidophiles thrive in acidic conditions (pH below 5.5)
(0.5mark) neutrophiles - prefer neutral conditions (pH 5.5-8.5)

(0.5mark) and alkaliphiles grow best in alkaline conditions (pH above 8.5) (0.5mark)

34. The spread plate method is a microbiological laboratory technique for isolating and counting the viable microorganisms present in a liquid sample by spreading a certain volume of the sample over an appropriate solidified culture media. (1.5mark). Following the incubation, there will be the formation of evenly distributed discrete colonies all over the surface of the culture media. This technique is used for isolating and counting the total number of viable microorganisms**.** (1.5mark)

35. Their ability to grow refrigeration temperature, antibiotic/disinfectants resistance, utilize a wide range of non carbohydrate carbon compounds, ability to produce variety of products, ability to use simple nitrogenous foods, ability to syntesize their own growth factors or vitamins, proteolytic and lipolytic activity, aerobic nature, pigment production etc (Any three points, 1mark can be given for each point)

36. Perishable foods – spoil readily unless special preservative methods are not used, example -meat, fish etc. Semi-perishable foods – If these foods are properly handled and stored they will remain unspoiled for a fairly long time. Example- apples, potatos etc

37. Synergistic effect, lower energy and chemical usage, improved safety and quality retention, extended shelf life, reduced need for chemical preservatives, flexible and adaptable, better sensory and nutritional qualities, consumer appeal etc (any four points - 0.75 mark for each point)

38. Selection and cleaning (0,5mark) peeling and cutting (0.5mark) blanching (0.5mark) filling cans 0.25 mark), sealing (0.25 mark), heat process (0.5 mark), cooling and storage (0.5 mark).

PART F

39. Definition of medium (1Mark), water, energy sources, nitrogen sources, minerals, buffers (if needed) etc, Supplemental components like growth factors, vitamins, antibiotics or other substances to support specific growth requirements etc. Soldifying agents .(1Mark)

Basal media (1Mark), enriched media (1Mark), enrichment media (1Mark), selective media (1Mark), differential media (1Mark), transport media(1Mark)

40. Drying brief introduction (2marks), Different types -solar drying (1.5marks), drying by mechanical dryer (1.5marks), drying during smoking (1.5marks), freeze drying (1.5marks).